CHEN40140: BIOPHARMACEUTICAL ENG. PROJECTS 25/26

MEngSc Biopharmaceutical Engineering

School of Chemical and Bioprocess Engineering, UCD

Introduction 3
Project 1: Optimisation of Plasmid DNA Yield Using a Commercial Preparation Kit 4
Project 2: Comparative Analysis and Design Space Mapping of Ultrafiltration Membranes for Protein Concentration 5
Project 3: Inhibition of cellular pathways for AAV production in HEK293 cells 6
Project 4: Characterization of a plasma pen device for biomedical applications 7
Project 5: Plasma-treated deep eutectic solvents as a novel approach for drug delivery 8
Project 6: The effect of oxidative stress on the cellular secretome 9
Project 7: Fate of Veterinary products in Biological Treatment systems 10
Project 8: Algal Omega-3 Production in Wastewater 11
Project 9: Application of Forward Osmosis technology in biopharmaceutical manufacturing 12
Project 10: Dual-targeted of fluoroquinolone/macrolide – composed lipid nanoparticles for Legionella treatment 13
Project 11: Nicotinamide nanoformulations for topical anti-inflammatory treatments 14
Project 12: Evolution of sup-tRNA therapeutics by site saturation mutagenesis 15
Project 13: Triplet codon reprogramming in mammalian cell factories 16
Project 14: Investigating RNA passive loading into exosomes for targeted therapeutic delivery across the blood-brain-barrier 17
Project 15: Screening Simple Deep Eutectic Solvents for Antimicrobial Activity and Antibiotic Enhancement to combat antimicrobial resistance 18
Project 16: Hybrid mixed metal Enzymatic Catalysis for Highly Stereoselective Synthesis of Chiral Alcohols 19
Project 17: In-vitro glycoengineering of CD16A 20
Project 18: BLI of glycoengineered CD16A 21
Project 19 Development of Glycan Profiling Method for IgGs 22
Project 20: Developing imaged capillary isoelectric focusing methods for charge variant profiling of complex biopharmaceuticals. 23
Project 21: Coupling Raman spectroscopy with size exclusion chromatography during downstream processing of monoclonal antibodies. 24
Project 22: Development of Advanced analytical methods for the quantification of mRNA stability 25
Project 23: Discovering Novel Proteins involved in rAAV5 Production through Machine Learning–Enhanced Proteomics 26
Project 24: Can machine learning based fragmentation prediction improve the detection of host cell proteins using mass spectrometry? 27

Project 25: Mining Novel Growth and Productivity Pathways: Machine Learning Analysis Proteomics data from Chinese Hamster Lung and Ovary Cell Factories 28

Project 26: Understanding the role of the CHO surfaceome 29

Project 27: Enhanced Mapping of the CHO cell Secretory Pathway 30

Project 28: Understanding the role of miR29a in CHO-K1 cells..... 31

Project 29: Understanding the role of miR29b-1 in CHO-K1 cells..... 32

Project 30: Determining the impact of IgG1 monoclonal antibody production in CHO cells 33

Project 31: Identification of transcriptome alteration associated with EPO-Fc production in CHO cells..... 34

Project 32: What impact do different media formulations have on CHO cell factories? 35

Project 33: Comparing the transcriptomes of CHO cell pools vs clonal populations ... 36

Project 34: Identification of mRNAs associated with lactate production during CHO cell culture..... 37

Project 35: Understanding the transcriptional basis for rapid CHO cell growth rate 38

Project 36: How does the biological system of emerging Chinese hamster cell lines compare to traditional CHO cells 39

Project 37: How does reducing cell culture temperature to 33 °C affect the CHO cell transcriptome? 40

Project 38: What is the biological basis of clonal variation in IgG1 production rates?... 41

Project 39: Comparing mRNA expression between the exponential and stationary phases for CHO-S cells 42

Project 40: Comparing high and low producing CHO cells 43

Project 41: Do subclones derived from existing cell lines have a different transcriptome? 44

Project 42: How does reducing cell culture temperature to 31 °C affect the CHO cell transcriptome? 45

Project 43: Comparison of cell lines with different levels of aggregation of bispecific antibody (bsAb1) 46

Project 44: Comparison of cell lines with different levels of aggregation of bispecific antibody (bsAb2) 47

Project 45: Comparison of bispecific antibody producing cell line to the original host cell line (bsAb1) 48

Project 46: Comparison of bispecific antibody producing cell line to the original host cell line (bsAb2) 49

Project 47: Understanding clonal variation in therapeutic sulfatase in high cell density cultures 50

Introduction

The CHEN40140 Biopharmaceutical Engineering Project (30 credits) runs for 12 weeks during the summer trimester.

The research projects will begin on the 18th May and will be submitted on the 18th August.

Students will be assessed via

1. A written document (90% of Marks)
2. Oral Presentation (10% of Marks)

This document provides details on the available research projects for Summer 2026. The research project is conducted either in the research labs run by the School Faculty and/or NIBRT PIs (see the School and NIBRT websites for more information).

Students are required to rank each project in terms of preference by Friday, 30th January 2026. All available projects provided here must be included in the ranked list (to be provided in January). An Excel template has been provided for this purpose, and the results must be submitted using the following link: https://forms.gle/ZVk5EuvipfKXZ4kJA

In cases where more than 1 student wishes to do a particular project and if there is a tie in their ranked preference list, the project will be allocated based on the student's Autumn GPA.

Attendance

Students allocated to an Experimental Research Project (#1-#22) must be present Monday through Friday during regular working hours (i.e., 9 am-5 pm) at either UCD or NIBRT.

Those students conducting an Omics Research project (#23-#47) are required to attend lectures and supervision sessions in person. Note in May/June, an Omics summer school. There will be no option to attend these sessions or to receive supervision remotely.

The summer school will combine theory lectures and hands-on sessions using Galaxy for bioinformatics workflows and dedicated proteomics software. Students will gain practical experience in running, interpreting and documenting end-to-end analyses.

Regards,

Colin Clarke

Project 1: Optimisation of Plasmid DNA Yield Using a Commercial Preparation Kit

Principal Investigator:	Jessica Whelan
Laboratory Location:	*UCD*
Modality	*Viral vectors*
Focus Area	*Upstream*
Project Description	The generation of sufficient quantities of high purity DNA plasmids are a significant challenge for rAAV production via triple transfection of HEK cell lines. The aim of this project is to **significantly improve the final yield and quality of a target plasmid DNA** using a specific commercial plasmid preparation kit. The optimization will involve a systematic investigation of the entire upstream and downstream process, focusing on key variables that influence DNA output. The student will systematically investigate the impact of critical factors such as **Competent Cell Strain, Bacterial Growth Medium, Growth Conditions (Temperature/Time), and Prep Kit Parameters**—on the final plasmid DNA yield and purity.
Anticipated project outcomes	Based on the analysis of the results an optimized protocol for the chosen kit will be developed.
Indicative skills and experience to be gained	Microbial culture, plasmid prep and analysis.

Project 2: Comparative Analysis and Design Space Mapping of Ultrafiltration Membranes for Protein Concentration

Principal Investigator:	Jessica Whelan
Laboratory Location:	*UCD*
Modality	*Recombinant protein*
Focus Area	*Downstream*
Project Description	The focus of this project will be on comparing the performance of different commercial ultrafiltration (UF) membranes for the concentration of a target protein, with a strong emphasis on mapping the process design space using Design of Experiments (DoE). The student will quantify and compare the performance metrics (e.g., flux, protein transmission/recovery, fouling rate) of up to three commercially available UF membranes (e.g., Polyethersulfone (PES), Regenerated Cellulose (RC), and Polytetrafluoroethylene (PTFE) or similar) with the same nominal Molecular Weight Cut-Off (MWCO) during concentration of a model protein, Bovine Serum Albumin (BSA). A DoE approach will be applied to systematically investigate the effect of key operating parameters (e.g., Transmembrane Pressure (TMP), Feed Flow Rate, and Initial Protein Concentration on the concentration performance for each membrane type.
Anticipated project outcomes	Based on the analysis of the DoE results a recommendation of the optimal membrane and operating conditions will be made.
Indicative skills and experience to be gained	Downstream processing, protein quantification, DoE

Project 3: Inhibition of cellular pathways for AAV production in HEK293 cells

Principal Investigator:	Niall Barron
Laboratory Location:	*NIBRT*
Modality	*Gene Therapy*
Focus Area	*Upstream*
Project Description	HEK293 cells are the main cell line used in production of AAV-based gene therapies. There is considerable interest in improving the yield and quality of viral vectors produced by these cells. As cells have inherent defence mechanisms to counter viral replication this project will seek to inhibit those defence mechanisms with a view to improving AAV synthesis.
Anticipated project outcomes	Data on the impact of select pathway inhibition in HEK cells on AAV yield
Indicative skills and experience to be gained	Cell culture, transfection, transduction, flow cytometry, fluorescent microscopy, western blot, ELISA, qPCR.

Project 4: Characterization of a plasma pen device for biomedical applications

Principal Investigator:	Daniela Boehm
Laboratory Location:	*UCD*
Modality	*Cell therapy*
Focus Area	*Upstream*
Project Description	Cold atmospheric plasma (CAP) is a partially ionized gas generated at (near) room temperature and atmospheric pressure which comprises a range of reactive species (particularly reactive oxygen and nitrogen species), along with electric field and UV radiation. The exposure of a liquid to plasma discharge results in the generation and retention of reactive oxygen and nitrogen species (RONS) within the liquid which can induce various effects when applied to cells, ranging from enhanced proliferation, to cell damage and death, depending on the composition and concentration of the reactive species. This range of cellular responses generates diverse application opportunities for cold plasma including in regenerative medicine and cancer treatment. Plasma treatment parameters determine the type and concentration of reactive species generated and the subsequent biological response of cells. This research will perform an in-depth characterization of a plasma pen device in relation to key plasma parameters such as frequency, amplitude and treatment time and apply a DoE approach across these parameters. The characterization of long-lived reactive species in plasma-treated liquids such as H2O2, nitrite and nitrate will be performed using spectrophotometric assays and the penetration depth of these species will be analysed in hydrogel models. Measurements of cell viability/proliferation of cell lines will be used to determine cell-toxic and cell-tolerable parameters for this device.
Anticipated project outcomes	Optimization of plasma treatment parameters and timing for cell-toxic and cell-tolerable applications.
Indicative skills and experience to be gained	cell culture, cell analysis, viability and metabolic activity assays, microscopy, colourimetric assays for chemical analyses, use of cold atmospheric plasma system, DoE

Project 5: Plasma-treated deep eutectic solvents as a novel approach for drug delivery

Principal Investigator:	Daniela Boehm
Laboratory Location:	*UCD*
Modality	*drug delivery*
Focus Area	*Fill finish (drug delivery)*
Project Description	Cold atmospheric plasma (CAP) is a partially ionized gas generated at (near) room temperature and atmospheric pressure which comprises a range of reactive species (particularly reactive oxygen and nitrogen species), along with electric field and UV radiation. The exposure of a liquid to plasma discharge results in the generation and retention of reactive oxygen and nitrogen species (RONS) within the liquid which can induce various effects when applied to cells, ranging from enhanced proliferation, to cell damage and death, depending on the composition and concentration of the reactive species. Deep eutectic solvents (DESs) are mixtures composed of two or more solid or liquid components that form a eutectic mixture with a lower melting point than the individual components. DESs may be used as solvents to improve the formulation of drugs where the active pharmaceutical ingredients (API) are poorly soluble in an aqueous medium. Plasma treatment of DESs provides an interesting opportunity to generate oxidative species within a DES and modulate its biological effects on cells, dependent on the properties of the DES and chemistry induced by the plasma. Applications may be in the drug delivery space with opportunities to improve drug solubility and/or activity as well as potential drug-plasma synergies. This research builds on the outcomes of a previous project where a range of DES were screened for their biological effect and changes induced by plasma treatment. Candidate DES compositions will be investigated in detail with regards to key chemical properties such as concentration of oxidative species and changes in chemical composition and biological effect post-plasma treatment. Different plasma treatment parameters including treatment time and discharge parameters of the plasma device will be evaluated and key plasma reactive species will be quantified using colorimetric assays. The cellular response to plasma-treated DESs will be assessed based on analyses such as metabolic activity, cell mass, colony-forming potential and oxidative stress.
Anticipated project outcomes	Optimization of plasma treatment parameters for the modulation of chemical properties and biological effects of deep eutectic solvents
Indicative skills and experience to be gained	Cell culture, cell analysis, viability assays, metabolic activity, microscopy, colourimetric assays for chemical analyses, use of cold atmospheric plasma system

Project 6: The effect of oxidative stress on the cellular secretome

Principal Investigator:	Daniela Boehm
Laboratory Location:	*UCD*
Modality	*Cell therapy*
Focus Area	*Analytical*
Project Description	Oxidative stress can cause damage to cell membranes and cellular components such as proteins or DNA and pose a threat to cell proliferation and survival. Cells respond to oxidative stress through various mechanism, including the upregulation of stress-responsive genes and anti-oxidant machinery. Cellular stress responses also impact cell-cell communication pathways via the cells' secretome, i.e. the release of cellular signalling factors and extracellular vesicles. Cold atmospheric plasma (CAP) is a partially ionized gas generated at (near) room temperature and atmospheric pressure which comprises a range of reactive species (particularly reactive oxygen and nitrogen species), along with electric field and UV radiation. Plasma can induce various effects on cells, ranging from enhanced proliferation, to cell damage and death, depending on treatment parameters such as treatment time or plasma power, voltage or frequency which impact the concentrations of reactive oxygen and nitrogen species (RONS) generated. Of particular interest to our group is understanding the mechanism by which plasma modulates cell behaviour, which may be used to inactivate cancer cells or enhance the expansion of difficult to grow cells such as stem cells. This research will investigate the effect of different levels of oxidative stress on the secretome of cells in culture. Conditioned media will be collected from cells that have been exposed to oxidative stress either through cold atmospheric plasma or chemicals such as H_2O_2 and analysed for a range of cytokines and cell signalling molecules. Conditioned medium may be fractionated into particular molecular size ranges and key molecules such as interleukins or TNF-α detected using analytical methods such as cytokine arrays and ELISA. Finally the effect of the conditioned medium on the growth and viability of other cells will be analysed using cell counting, metabolic activity and cell mass determination. Insight into secretome changes with oxidative stress may help to understand cellular stress responses and be used to improve cell expansion strategies, particularly of primary cells for cell therapy applications.
Anticipated project outcomes	Identification of key factors secreted by cells exposed to oxidative stress and their effects on target cells.
Indicative skills and experience to be gained	Protein fractionation, cytokine arrays, ELISA, Cell culture, cell analysis, viability assays, metabolic activity, microscopy, use of cold atmospheric plasma system

Project 7: Fate of Veterinary products in Biological Treatment systems

Principal Investigator:	Dr Eoin Syron
Laboratory Location:	*UCD*
Modality	*Biological Degradation of APIs*
Focus Area	*Environmental protection*
Project Description	As we transition to a circular economy the reuse of waste products for value recovery is becoming more viable. However municipal and industrial wastewaters are contaminated by a wide range of chemicals,(Micropollutants) including pharmaceutical residues, personal care products, various household chemicals, and biocides/pesticides. This is similar animal manures which are contaminated with veterinary products. In his project we will investigate the fate of these Micropollutants in treatment systems that are recovering value e.g. nutrients and energy from the animal manure. Are these products broken down, do they impact the recovery processes, are they released into the environment, where they may generate adverse effects on aquatic organisms. This all depends on their fate in treatment system.
Anticipated project outcomes	Running two biological treatment reactors to produce fertilisers. Use of filtration equipment, membrane process and electrocoagulation for the separation of suspended and dissolved solids.
Indicative skills and experience to be gained	Analytical techniques, Bioreactor operation, Membrane separation process-literature review

Project 8: Algal Omega-3 Production in Wastewater

Principal Investigator:	Dr Eoin Syron
Laboratory Location:	*UCD*
Modality	*Micro-organisim growth rates for enhanced*
Focus Area	*i.e. Upstream/Downstream/Fill Finish/Analytics*
Project Description	**Algal Omega-3 Production in Wastewater** This project investigates a synergistic, circular-economy model: using microalgae to simultaneously remediate nutrient-polluted wastewater while cultivating a valuable biomass feedstock for omega-3 fatty acids. The central challenge is process optimization. Algal growth in variable wastewater is unpredictable, and peak omega-3 production often occurs under nutrient stress, which may not align with peak biomass. Traditional, slow monitoring methods make real-time adjustments impossible. We propose using in-vivo chlorophyll-a fluorescence as a rapid, non-invasive monitoring tool. We hypothesize that fluorescence signals can not only provide a reliable proxy for algal biomass but also act as a sensitive indicator of the algae's physiological state. We expect these "stress" signatures to correlate directly with the onset of high-level omega-3 accumulation. By cultivating *Nannochloropsis sp.* on treated wastewater and a control medium, we will correlate fluorescence data with biomass, nutrient depletion, and omega-3 content.
Anticipated project outcomes	The expected outcome is a validated model that uses real-time fluorescence data to identify the precise optimal harvest time, maximizing omega-3 yield from a waste stream
Indicative skills and experience to be gained	Algal growth, Analytical techniques, process intensification

Project 9: Application of Forward Osmosis technology in biopharmaceutical manufacturing

Principal Investigator:	Dr Eoin Casey
Laboratory Location:	*UCD*
Modality	*Recombinant protein*
Focus Area	*Downstream*
Project Description	This project will investigate the potential application of forward osmosis technology in biopharmaceutical manufacturing, specifically in the area of downstream processing (DSP) . DSP involves separation, extraction, concentration, purification and packaging of the product after fermentation. There are many challenges associated with DSP and its technical complexity means that its costs account for a large fraction of the total manufacturing costs of major biopharmaceutical products. Many separation processes require elevated pressure or shear stress which can damage the complex therapeutic product. Therefore, investigations into novel DSP technologies are important to help improve performance and lower costs. Forward Osmosis (FO) is not currently applied in biopharmaceutical manufacturing and this project will assess its potential for certain specialised applications. In FO, two liquids of differing concentrations are separated by a membrane. The difference in concentration results in permeation of pure water from the low concentration liquid to the high concentration liquid. Thus, one liquid is diluted whilst the other is concentrated. It eliminates the high temperature and pressure requirements of conventional concentration and dilution techniques. Tangry, V., Haddad, W., Behboudi, A. and Zydney, A.L., 2025. The application of forward osmosis for producing highly concentrated biotherapeutics. Journal of Membrane Science, 721, p.123839. https://doi.org/10.1016/j.memsci.2025.123839
Anticipated project outcomes	In this project, opportunities for FO in DSP will be identified and investigated experimentally and the student will need to do some basic mathematical modelling as part of the data analysis.
Indicative skills and experience to be gained	Purification processes, experimental and modelling

Project 10: Dual-targeted of fluoroquinolone/macrolide – composed lipid nanoparticles for Legionella treatment

Principal Investigator:	Eliana Souto (eliana.souto@ucd.ie)
Laboratory Location:	*UCD*
Modality	*Drug delivery*
Focus Area	*Top-down production of targeted delivery systems & analytics for quality control*
Project Description	Production, by top-down methodology, of solid lipid nanoparticles (SLN) containing a selected macrolide and nanostructured lipid carriers (NLC) containing selected fluoroquinolone, both commonly used in the treatment of legionella. The freshly prepared drug-loaded SLN and NLC will be physico-chemically characterized (i.e., particle size parameters, zeta potential, polymorphism, rheology) and compared to those stored at 4°C and 22°C for a minimum of 30 days. Mathematical modeling (i.e., Higuchi, zero order, first order, Korsmeyer-Peppas) will be done to describe the release profile in simulated gastrointestinal (biorelevant) media. Development of a gastro-resistant solid dosage form for the dual delivery of the antibiotics-loaded into LNPs, and its characterization following USP and/or EP guidelines.
Anticipated project outcomes	Two research articles.
Indicative skills and experience to be gained	Domain of: Top-down production of lipid nanoparticles. Design of experiments. Mathematical modelling of release kinetics. Gastro-resistant solid dosage forms.

Project 11: Nicotinamide nanoformulations for topical anti-inflammatory treatments

Principal Investigator:	Eliana Souto (eliana.souto@ucd.ie)
Laboratory Location:	*UCD*
Modality	*Drug delivery*
Focus Area	*Top-down production of targeted delivery systems & analytics for quality control*
Project Description	In this project, it is proposed the top-down production of nanostructured lipid carriers (NLC), composed of glycerol behenate and long-chain triglycerides, for the loading of nicotinamide (vitamin B3) to exploit its topical-anti-inflammatory properties and absorption enhancer capacity of lipid nanoparticles. The freshly prepared drug-loaded NLC will be physico-chemically characterized (i.e., particle size parameters, zeta potential, polymorphism, rheology) and compared to those stored at 4°C and 22°C for a minimum of 30 days. Release profile of loaded drug from NLC will be done in vertical Franz diffusion cells using synthetic membranes to simulate skin barrier, and their release kinetics fit to different models (i.e., Higuchi, zero order, first order, Korsmeyer-Peppas). Development of a semi-solid product for topical administration and its characterization (rheology and release) following USP and/or EP guidelines.
Anticipated project outcomes	Two research articles.
Indicative skills and experience to be gained	Domain of: Top-down production of lipid nanoparticles. Design of experiments. Mathematical modelling of release kinetics. Rheology of semi-solids.

Project 12: Evolution of sup-tRNA therapeutics by site saturation mutagenesis

Principal Investigator:	Alan Costello
Laboratory Location:	*NIBRT*
Modality	*Gene therapy*
Focus Area	*Upstream*
Project Description	Premature termination codons (PTCs) are responsible for ~10–15% of all genetic disease. PTC mutations inhibit synthesis of full-length proteins, and may be more deleterious than missense mutations as they result in the loss of protein expression. PTCs are associated with many severe disease phenotypes, including: cystic fibrosis, Duchenne muscular dystrophy, cancer, and more. Suppression of a PTC through competition with or inhibition of cellular translation release factors would in theory restore translation of full-length protein and normal cell function. Transfer RNA (tRNA)-based therapeutics show promise as precision medicines to treat ~7000 known rare genetic diseases arising from PTCs. tRNA genes can be engineered by altering the base-pairing of the anticodon to correspond to a PTC mutation. These engineered tRNAs are referred to as sup-tRNAs. One potential issue with sup-tRNA therapy relates to readthrough of native termination codons. This project will combine guide small nucleolar RNAs (gsnoRNAs) to programme pseudouridylation of PTCs and directed evolution of sup-tRNAs by site saturation mutagenesis. Through this you will be screening for sup-tRNA mutants with a bias for PTCs readthrough over native termination codons. For background on tRNA therapeutics and tRNA evolution, see: Liu, J., Yan, X., Wu, H. et al. RNA codon expansion via programmable pseudouridine editing and decoding. Nature 643, 1410–1420 (2025). https://doi.org/10.1038/s41586-025-09165-x Costello, A., Peterson, A.A., Lanster, D.L. et al. Efficient genetic code expansion without host genome modifications. Nat Biotechnol (2024). https://doi.org/10.1038/s41587-024-02385-y
Anticipated project outcomes	Evaluation of mRNA modification by snoRNAs Screen site saturation mutagenesis library of sup-tRNAs
Indicative skills and experience to be gained	Experimental design Synthetic biology design-build-test-learn paradigm DNA assembly and cloning In vitro biochemistry Mammalian cell culture

Project 13: Triplet codon reprogramming in mammalian cell factories

Principal Investigator:	Alan Costello
Laboratory Location:	*NIBRT*
Modality	*Recombinant protein*
Focus Area	*Upstream*
Project Description	Natural proteins often require post-translational modifications (PTMs) to fulfil their structural and functional diversity. PTMs such as phosphorylation, glycosylation, methylation, acetylation, ubiquitination, and others are critical for regulating protein activity, localization, stability, and interactions. Unfortunately PTM profiles can be variable between cell types and cause issues with recombinant product quality. By expanding the genetic code to incorporate noncanonical amino acids (ncAAs) which mimic natural PTMs, we may bypass heterogeny in post-translational modification and improve product quality profiles. Several strategies exist to expand the genetic code; stop codon suppression, reassigning synonymous sense codons, altering codon size, or expanding the genetic alphabet. In this project you will build a DNA construct encoding a genetic code expansion circuit. You will use this circuit to monitor genetic code expansion activity in mammalian cells and compare the ncAA incorporation efficiency at stop codons and low usage sense codons. Finally, site-specific ncAA incorporation will be validated using mass spectrometry. For background on genetic code expansion, see: Costello, A., Peterson, A.A., Chen, P.H., Bagirzadeh, R., Lanster, D.L. and Badran, A.H., Genetic code expansion history and modern innovations. Chemical Reviews (2024). https://pubs.acs.org/doi/full/10.1021/acs.chemrev.4c00275 Ding, W., Yu, W., Chen, Y., Lao, L., Fang, Y., Fang, C., Zhao, H., Yang, B. and Lin, S., 2024. Rare codon recoding for efficient noncanonical amino acid incorporation in mammalian cells. Science, 384(6700), pp.1134-1142. https://www.science.org/doi/10.1126/science.adm8143
Anticipated project outcomes	Design and test a genetic code expansion circuit in a mammalian cell line Evaluate relative efficiency of low usage sense or stop codon decoding Validate ncAA incorporation by mass spectrometry
Indicative skills and experience to be gained	Experimental design Synthetic biology design-build-test-learn paradigm DNA assembly and cloning Mammalian cell culture Mass spectrometry

Project 14: Investigating RNA passive loading into exosomes for targeted therapeutic delivery across the blood-brain-barrier

Principal Investigator:	Mark Smales
Laboratory Location:	*NIBRT*
Modality	*Exosomes*
Focus Area	*Upstream*
Project Description	Exosomes are naturally derived extracellular vesicles that have recently attracted much attention as vehicles to deliver therapeutic payloads to systemic and tissue specific targets. However, the loading, characterisation and targeting of exosomes across the blood brain barrier (BBB) remain a challenge. We have developed protocols to manufacture exosomes from cultured mammalian cell systems and chemically modify these to extend their half-life and aid with targeting across the BBB. This project will investigate enhanced manufacturing methods and subsequently different passive loading approaches to maximise payload incorporation into exosomes. We will particularly focus on loading RNA therapeutics including microRNAs and longer coding RNAs into exosomes and then evaluate, in a model of the BBB, the ability of differently loaded and decorated exosomes to deliver the payload into cells and subsequent impact on the target. Exosomes are naturally derived extracellular vesicles that have recently attracted much attention as vehicles to deliver therapeutic payloads to systemic and tissue specific targets. However, the loading, characterisation and targeting of exosomes across the blood brain barrier (BBB) remain a challenge. We have developed protocols to manufacture exosomes from cultured mammalian cell systems and chemically modify these to extend their half-life and aid with targeting across the BBB. This project will investigate enhanced manufacturing methods and subsequently different passive loading approaches to maximise payload incorporation into exosomes. We will particularly focus on loading RNA therapeutics including microRNAs and longer coding RNAs into exosomes and then evaluate, in a model of the BBB, the ability of differently loaded and decorated exosomes to deliver the payload into cells and subsequent impact on the target.
Anticipated project outcomes	Data on the delivery of exosomes across the BBB and potential efficacy
Indicative skills and experience to be gained	Cell culture, exosomes, fluorescent microscopy, western blot, ELISA, qPCR.

Project 15: Screening Simple Deep Eutectic Solvents for Antimicrobial Activity and Antibiotic Enhancement to combat antimicrobial resistance

Principal Investigator:	Roderick Jones
Laboratory Location:	*UCD*
Modality	
Focus Area	
Project Description	This project will investigate whether simple, easy-to-make deep eutectic solvents (DESs) can inhibit bacterial growth and enhance the activity of a common antibiotic to combat the rise of antimicrobial resistant pathogens. A small panel of DESs, prepared from biocompatible components such as choline chloride with glycerol, urea, or lactic acid, will be tested against safe laboratory strains (e.g. E. coli K-12 and a non-pathogenic Gram-positive bacterium) using broth microdilution and agar diffusion assays to assess antimicrobial effects. The most promising DES will then be combined with a standard antibiotic (e.g. ampicillin) to see if it increases the antibiotic's effectiveness in simple disc diffusion or microplate assays. Together, the results will provide proof-of-concept data on the potential of DESs as both direct antimicrobial agents and antibiotic potentiators.
Anticipated project outcomes	
Indicative skills and experience to be gained	

Project 16: Hybrid mixed metal Enzymatic Catalysis for Highly Stereoselective Synthesis of Chiral Alcohols

Principal Investigator:	Roderick Jones
Laboratory Location:	*UCD*
Modality	
Focus Area	
Project Description	This project aims to develop a hybrid catalytic system that combines the broad reactivity of a single-atom iron catalyst with the exquisite stereoselectivity of engineered enzymes to produce enantiomerically pure alcohols from simple hydrocarbons. First, an Fe–N–C single-atom catalyst will be synthesized and optimized to selectively oxidize ethylbenzene into acetophenone under mild, green conditions. This chemically generated intermediate will then be transformed into either the (R)- or (S)-enantiomer of 1-phenylethanol using commercially available alcohol dehydrogenases supported by a cofactor regeneration system. By integrating heterogeneous single-atom catalysis with biocatalysis in a sequential one-pot process, the project demonstrates a sustainable, modular approach to performing highly stereoselective chemical transformations that neither catalyst class can achieve alone
Anticipated project outcomes	Data on the impact of select pathway inhibition in HEK cells on AAV yield
Indicative skills and experience to be gained	

Project 17: In-vitro glycoengineering of CD16A

Principal Investigator:	Ioscani Jimenez Del Val / Yuan Feng
Laboratory Location:	UCD
Modality	Therapeutic glycoproteins and their CD16A receptor
Focus Area	Glycosylation & bioanalytics
Project Description	CD16A (FcγRIIIa) is an activating IgG receptor expressed on human NK cells and macrophages, mediating ADCC through interaction with the Fc region of IgG, which is central to the oncolytic activity of a multitude of commercial mAbs. The glycosylation of both the mAb Fc and CD16A critically influences their binding affinity, and CD16A exhibits inter-individual glycosylation heterogeneity. Developing methods to precisely control CD16A glycoforms will enable a more accurate representation of the N-glycan diversity of immune cells and, thereby, facilitate the creation of affinity maps between specific CD16A and IgG glycovariants. In turn, such affinity maps could enable personalisation of immunotherapeutics based on the glycosylation status of the patient. This project aims to test glycosidase kinetics on the *in vitro* glycan remodelling (glycoengineering) of CD16A. Commercial or CHO-K1 produced CD16A will be used, the CHO-K1 produced CD16A will be purified using IMAC FPLC. Targeted glycosidase treatments – sialidase (α3 and α6) and galactosidase (β4) – will be applied to modify specific terminal glycan residues and detect glycoform changes using a fluorescent lectin assay.
Anticipated project outcomes	Optimized lectin-plate assay for In-vitro glycoengineering of CD16A. Kinetic parameters (semi-quantitative) of glycosidase activity.
Indicative skills and experience to be gained	Expression and purification of recombinant proteins. Enzymatic reactions. Glycosylation regulation. Lectin-based plate assay development. Data analysis and visualization, experimental design, reporting.

Project 18: BLI of glycoengineered CD16A

Laboratory Location:	004B
Modality	Therapeutic glycoproteins and their CD16A receptor
Focus Area	Glycosylation & bioanalytics
Project Description	The FcγRIIIA (CD16A) receptor mediates ADCC, which is central to the oncolytic activity of a multitude of commercial mAbs. Thus, higher binding affinity between the mAb Fc and FcγRIIIA elicits increased ADCC and, therefore, increased oncolytic activity in patients. CD16A N-glycosylation is highly variable across the human population and varies with disease progression. Furthermore, CD16A N-glycosylation heavily influences the binding affinity with the mAb Fc. The binding affinity between mAbs and FcRs, measured as the affinity coefficient (Ka), can be measured using bioanalytical tools, such as biolayer interferometry (BLI) or surface plasmon resonance. This project will (i) treat purified WT CD16A with sialidase (α3, α6) and galactosidase (β4) to achieve near-complete removal of terminal sialic acid and galactose residues. (ii) Immobilize glycoengineered CD16A on BLI sensors. (iii) Measure binding kinetics (Ka, Kd, KD) of the modified CD16A with fucosylated and afucosylated mAbs using the Sartorius Octet BLI platform.
Anticipated project outcomes	An optimized method for *in vitro* sialidase and galactosidase glycoengineering. *In vitro* glycoengineered CD16A product. Ka measurement of IgG:FcγRIIIA interactions for a panel of glycoengineered CD16A and fucosylated and afucosylated mAb.
Indicative skills and experience to be gained	Understanding and optimization of enzymatic *in vitro* glycoengineering. Wet lab experience, sample preparation and handling, experimental design and method optimisation. Understanding and operation of a commonly used bioanalytical instrument, the Sartorius Octet N1.

Project 19 Development of Glycan Profiling Method for IgGs

Principal Investigator:	Ioscani Jiménez del Val/Sheryl Lim Li Yan
Laboratory Location:	UCD – 004B
Modality	CHO Cells/Recombinant Protein
Focus Area	Upstream/Analytics
Project Description	This project aims to develop a rapid, in-house glycan analysis workflow to characterise IgG products produced by inducible CHO cell lines (VRC01_dLin and NIST_dLin). The first phase will focus on optimising a glycan release and staining assay for IgG molecules with glycosylated Fab regions. This will enable comparison of fucosylation and galactosylation patterns under different induction conditions. In the second phase, the project will extend the analysis to the Fc region glycans, developing or adapting a method to access and profile Fc-linked glycans. Once validated, the new workflow will be used to compare glycan compositions and benchmarked using existing or archived IgG product glycan profiles. This work will contribute to establishing a cost-effective, internal screening method for analysing IgG glycosylation prior to outsourcing full glycan analyses. It will also generate useful preliminary data linking glycan composition to binding kinetics (via BLI) and supporting future process optimisation and product quality control.
Anticipated project outcomes	A validated in-house method for comparative Fab and Fc glycan analysis. Preliminary data on fucosylation and galactosylation trends in inducible CHO systems. Correlation between induction conditions and IgG glycosylation patterns. Foundational workflow to reduce reliance on outsourced glycan analysis and associated costs.
Indicative skills and experience to be gained	Mammalian cell culture and recombinant protein production. Protein purification techniques (e.g., Protein A affinity chromatography). Glycan release and analytical characterisation methods. Data interpretation in relation to bioprocess parameters and product quality attributes. Experimental design and optimisation for analytical method development.

Project 20: Developing imaged capillary isoelectric focusing methods for charge variant profiling of complex biopharmaceuticals.

Principal Investigator:	Jonathan Bones
Laboratory Location:	NIBRT
Modality	Recombinant Proteins
Focus Area	Analytics
Project Description (max. 250 words)	Posttranslational modification of recombinant proteins can result in alterations in the local surface charge distribution of the molecule leading to the formation of different charge variants. Traditionally, methods including cation exchange chromatography and isoelectric focusing are used for the determination of the charge variant profile of the molecule, a requirement under the ICH guidelines. This project will explore the development of charge variant separation methods using the Bio-Techne Maurice Flex instrument. Maurice Flex uses imaged capillary isoelectric focusing (icIEF) to separate and detect different charge variants present in biopharmaceutical samples. icIEF is a simple method and accordingly, has found widespread adoption within the quality control environment in the biopharmaceutical industry. While the method is well established for molecules such as monoclonal antibodies, this project will focus on the development of icIEF methods for more complex biopharmaceuticals such as multi-specific antibodies, Fc fusion proteins or antibody drug conjugates. Optimisation of separation conditions will be performed to ensure maximal separation of charge variants present. Following optimisation of separation conditions, fraction collection will then be performed and verified using the fractionating chip capabilities of Maurice Flex. Collected fractions will then be subjected to peptide mapping or intact mass analysis to determine the posttranslational modifications present and responsible for the formation of each specific charge variant detected. To further understand the observed selectivity, various modelling methods will be applied.
Anticipated project outcomes	1. Optimised sample preparation of different complex biopharmaceuticals for icIEF separation using Maurice Flex. 2. Development of high performance charge variant separation methods for charge variant profiling using icIEF. 3. Development of efficient fractionation methods for collection of charge variants and verification of the separation performance. 4. Identification of posttranslational modifications responsible for the formation of observed charge variants.
Indicative skills and experience to be gained	Students will gain experience in: 1. Sample preparation strategies, material handling and experimental design. 2. Utilisation of state-of-the-art analytical instrumentation. 3. Working within a diverse team to achieve their research objectives. 4. Effectively communicating and reporting their experimental outcomes to various stakeholders.

Project 21: Coupling Raman spectroscopy with size exclusion chromatography during downstream processing of monoclonal antibodies.

Principal Investigator:	Jonathan Bones
Laboratory Location:	NIBRT
Modality	Recombinant Proteins
Focus Area	Analytics
Project Description (max. 250 words)	Raman spectroscopy has attracted considerable attention over the last decade with significant adoption as a process analytical technology for upstream process monitoring. Despite its adoption, Raman is not a direct method, requiring significant data processing to enable the detection of useful analytical features from which predictive models can be determined. This limitation can be partly attributed to the high aqueous background and relatively the low concentration of analytes during an upstream process regardless of the scale used. This project will investigate the application of Raman spectroscopy during downstream processing of monoclonal antibodies. Spectra will be collected in flow through mode by the application of a flow cell, to which the Raman probe can be directly attached, post affinity or size based separation. The project will investigate concentration sensitivity and will look to find and identify spectral features present that correlate with protein concentration and ideally product quality attributes. The sensitivity of the method will also be investigated, considering the concentrating effect that the protein experiences during downstream processing and subsequent water elimination, it is hoped that sensitivity and information content will increase accordingly relative to what can be determined during upstream process modelling. The associated student will also be develop associated models based on the collected spectral data that can be deployed for monitoring protein concentration or quality attributes using the Raman setup. Where possible, verification using orthogonal data generation will be performed.
Anticipated project outcomes	5. Creation of a flow through Raman set up for deployment during downstream processing of monoclonal antibodies. 6. Investigate system performance and information content. 7. Development of multivariate models using collected spectral data that enable inline measurement of quality attributes. 8. Verification of performance using orthogonal methods.
Indicative skills and experience to be gained	Students will gain experience in: 5. Analytical and purification strategies, material handling and experimental design. 6. Utilisation of state-of-the-art analytical instrumentation. 7. Working within a diverse team to achieve their research objectives. 8. Effectively communicating and reporting their experimental outcomes to various stakeholders.

Project 22: Development of Advanced analytical methods for the quantification of mRNA stability

Principal Investigator:	Steven Ferguson
Laboratory Location:	NIBRT
Modality	mRNA
Focus Area	*Downstream*
Project Description	mRNA vaccines have revolutionized the treatment of infectious diseases. However outside of vaccine applications they have yet to deliver on their vast potential. One of the key limitations for protein replacement, gene therapy and cell therapy applications is their limited stability. Quantifying the breakdown of the native and modified mRNA under biologically relevant conditions provides a means to optimize mRNA products in this regard. However one limitation is the lack of platforms that have both sufficient nucleotide level resolution and the ability to quantify in unbiased fashion the degradation pathways for these molecules.
Anticipated project outcomes	Data on mRNA instability
Indicative skills and experience to be gained	A mixture of novel nanopore sequencing and mass spectrometry techniques will be used to bridge this cap in capability and complement on-going mRNA chemical modification workflows in the Ferguson research group. The student will work with this team on the development and application of these analytical methods with a focus on formulation stability.

Project 23: Discovering Novel Proteins involved in rAAV5 Production through Machine Learning–Enhanced Proteomics

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	Understanding cell factory biology is crucial for advancing AAV based gene therapy manufacturing. Mass spectrometry–based proteomics can reveal the biology underlying desirable HEK293 cell manufacturing performance, but interpreting this data is challenging. This project will explore how new approaches in machine learning can improve can improve protein identification. The project will aim to uncover new insights into how HEK293 cell factories function, adapt, and could be engineered in the future for high-efficiency gene therapy production. This project will apply state-of-the-art machine learning models for peptide fragmentation prediction to improve interpretation of mass spectrometry data from rAAV5 production. By training deep learning models on large peptide spectral libraries, these tools enable highly accurate predictions of fragmentation spectra for both known and novel peptides. Integrating such machine learning predictions enhances confidence in protein identification, uncovers previously undetectable proteins, and enables deeper proteome coverage. This approach has the potential to generate new biological insights from proteins found to be differentially expressed during AAV production Strasser L, Boi S, Guapo F, Donohue N, Barron N, Rainbow-Fletcher A, Bones J. Proteomic Landscape of Adeno-Associated Virus (AAV)-Producing HEK293 Cells. Int J Mol Sci. 2021 Oct 25;22(21):11499. doi: 10.3390/ijms222111499. PMID: 34768929; PMCID: PMC8584267
Anticipated project outcomes	The identification of novel proteins involved in rAAV5 production
Indicative skills and experience to be gained	Students will learn how to use advanced machine learning tools (such as Prosit and AlphaPeptDeep) to generate in silico fragmentation spectra, gaining practical experience in training, evaluating, and applying deep learning models to real-world proteomic datasets. They will develop skills in spectral matching and peptide identification, which are critical for increasing confidence in protein detection. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** While this is a distinct project the student will work as part of a wider Team (see: projects 24, 25, 26, 27) all focused on proteomic bioinformatics.

Project 24: Can machine learning based fragmentation prediction improve the detection of host cell proteins using mass spectrometry?

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project aims to identify and characterize host cell proteins (HCPs) present in monoclonal antibody (mAb) drug products by combining advanced mass spectrometry–based proteomics with machine learning peptide fragmentation prediction. Students will process and analyze proteomics data from mAb production and purification workflows. By leveraging modern machine learning models that predict peptide fragmentation spectra, this approach will increase the accuracy and depth of peptide and protein identification—enabling the detection of low-abundance and novel HCPs often missed by conventional pipelines. The results will be used to map the persistence, origins, and removal trajectories of individual HCPs throughout mAb bioprocessing. Falkenberg H, Waldera-Lupa DM, Vanderlaan M, Schwab T, Krapfenbauer K, Studts JM, Flad T, Waerner T. Mass spectrometric evaluation of upstream and downstream process influences on host cell protein patterns in biopharmaceutical products. Biotechnol Prog. 2019 May;35(3):e2788. doi: 10.1002/btpr.2788. Epub 2019 Mar 6. PMID: 30767403.
Anticipated project outcomes	Demonstration of more confident identification of host cell proteins—especially those of low abundance within the final mAb drug product.
Indicative skills and experience to be gained	Core principles of mAb manufacturing, host cell protein risks, and regulatory concerns. Hands-on experience applying machine learning models for in silico peptide fragmentation, understanding how AI augments protein/peptide identification. Skills in advanced proteomics software, data visualization, and troubleshooting challenges specific to HCP detection. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** While this is a distinct project the student will work as part of a wider Team (see: projects 23, 25, 26, 27) all focused on proteomic bioinformatics.

Project 25: Mining Novel Growth and Productivity Pathways: Machine Learning Analysis Proteomics data from Chinese Hamster Lung and Ovary Cell Factories

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project aims to uncover new biological strategies for improving monoclonal antibody (mAb) production by analyzing proteomics data from both Chinese hamster lung (CHL-YN) cells and the industry-standard Chinese hamster ovary (CHO) cells. Using state-of-the-art machine learning methods, students will analyze mass spectrometry–based proteomic profiles to identify unique proteins, pathways, or post-translational modifications that drive rapid growth, energy metabolism, or product yield in CHL-YN cells. The goal is to discover transferable traits or regulatory mechanisms that could inform next-generation engineering of CHO cells, expanding options for high-performing biopharmaceutical manufacturing hosts. Tsunoda Y, Arishima R, Boronina T, Cole R, Yamano-Adachi N, Betenbaugh M, Omasa T. Comparative omics profiling reveals differences in biomass, energy production, and vesicle transport between CHO and fast-growing CHL-YN cells. Sci Rep. 2025 Nov 13;15(1):39784. doi: 10.1038/s41598-025-23503-z. PMID: 41233401; PMCID: PMC12615816.
Anticipated project outcomes	Identification of distinct or enhanced biological features in CHL-YN cells that contribute to superior cell growth and protein biosynthesis rates compared to CHO cells. Discovery of new candidate proteins or pathways that could be engineered or regulated to improve productivity and efficiency in CHO cell factories. Generation of actionable biological hypotheses for future cell line development, supported by machine learning–driven analysis of large-scale proteomics data
Indicative skills and experience to be gained	Application of machine learning techniques in analyzing omics datasets to detect complex biological differences and identify engineering targets. Skills in integrating biological data from different cell types and tissues to contextualize findings for real-world manufacturing relevance. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** While this is a distinct project the student will work as part of a wider Team (see: projects 23, 24, 26, 27) all focused on proteomic bioinformatics.

Project 26: Understanding the role of the CHO surfaceome

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	Upstream
Project Description	This project will explore the cell surface protein landscape (surfaceome) of Chinese hamster ovary (CHO) production cells to reveal new targets for engineering scalable and robust biomanufacturing hosts. By applying machine learning to mass spectrometry–based proteomics data—including peptide fragmentation prediction—students will interpret large-scale surfaceome datasets to identify proteins and pathways associated with cell adhesion, aggregation, and microenvironmental response. The focus will be on uncovering actionable biological features, such as key adhesion molecules or receptors, that mediate undesirable cell aggregation or signal integration in production environments. The integration of predictive machine learning models will increase confidence in mapping complex cell surface networks, enabling the discovery of both known and novel factors that can be targeted to reduce aggregation and improve cell growth, viability, and protein yields. Klingler F, Mathias S, Schneider H, Buck T, Raab N, Zeh N, Shieh YW, Pfannstiel J, Otte K. Unveiling the CHO surfaceome: Identification of cell surface proteins reveals cell aggregation-relevant mechanisms. Biotechnol Bioeng. 2021 Aug;118(8):3015-3028. doi: 10.1002/bit.27811. Epub 2021 May 14. PMID: 33951178.
Anticipated project outcomes	Identification of surfaceome features, including receptors, transporters, and adhesion proteins. Generation of biological hypotheses that guide rational engineering of the cell surface to reduce aggregation and enhance scalability in mAb production processes. Refined workflows for the use of machine learning–assisted proteomics in complex cell surface protein datasets.
Indicative skills and experience to be gained	The significance of the CHO cell surfaceome for biomanufacturing and strategies for targeted host cell engineering. Hands-on experience with machine learning models for protein identification in proteomics. Competence in integrating functional annotation, pathway enrichment, and experimental validation for engineering recommendations.Critical thinking to translate large, multi-dimensional protein datasets into biological action plans addressing industry-relevant challenges. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** While this is a distinct project the student will work as part of a wider Team (see: projects 23, 24, 25, 27) all focused on proteomic bioinformatics.

Project 27: Enhanced Mapping of the CHO cell Secretory Pathway

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use machine learning to analyze subcellular proteomics data, focusing on the classical secretory pathway (CSP) in Chinese hamster ovary (CHO) production cells. Students will examine global protein expression changes across CHO cell lines with differing monoclonal antibody productivities, using peptide fragmentation prediction and quantitative proteomics tools. Machine learning approaches will be applied to classify and prioritize hundreds of differentially expressed CSP proteins, uncovering new patterns and pinpointing previously overlooked molecular bottlenecks or regulatory features. The goal is to provide actionable insights into the specific proteins and the processes that can be engineered or modulated to relieve production bottlenecks and enhance recombinant protein yields. Pérez-Rodriguez S, Wulff T, Voldborg BG, Altamirano C, Trujillo-Roldán MA, Valdez-Cruz NA. Compartmentalized Proteomic Profiling Outlines the Crucial Role of the Classical Secretory Pathway during Recombinant Protein Production in Chinese Hamster Ovary Cells. ACS Omega. 2021 May 3;6(19):12439-12458. doi: 10.1021/acsomega.0c06030. PMID: 34056395; PMCID: PMC8154153.
Anticipated project outcomes	Discovery and annotation of novel CSP components and associated pathways linked to high or low protein production phenotypes. Identification of key proteins associated with ER and Golgi function, vesicle trafficking, protein folding, glycosylation, and secretion that form new engineering targets. Generation of a prioritized list of candidate proteins and pathways for future cell line engineering, informed by machine learning–driven proteome analysis.
Indicative skills and experience to be gained	Application of machine learning algorithms for classification, clustering, and ranking of differentially expressed proteins in omics data. Practical skills in integrating computational proteomics tools for peptide identification, quantification, and biological annotation. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** While this is a distinct project the student will work as part of a wider Team (see: projects 23, 24, 25, 27) all focused on proteomic bioinformatics.

Project 28: Understanding the role of miR29a in CHO-K1 cells

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will reanalyse the RNA-seq data to define the specific transcriptomic signature and pathways regulated by miR-29a in CHO-K1 cells. The students will compare anti-miR and pre-miR conditions to identify miR-29a–sensitive genes and enriched pathways. Muluhngwi P, Richardson K, Napier J, Rouchka EC, Mott JL, Klinge CM. Regulation of miR-29b-1/a transcription and identification of target mRNAs in CHO-K1 cells. Mol Cell Endocrinol. 2017 Mar 15;444:38-47. doi: 10.1016/j.mce.2017.01.044. Epub 2017 Jan 28. PMID: 28137615; PMCID: PMC5316361.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 29: Understanding the role of miR29b-1 in CHO-K1 cells

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will focus on miR-29b-1–mediated regulation, using the RNA-seq dataset to characterise common and unique gene targets and signalling pathways relative to miR-29B-1. Muluhngwi P, Richardson K, Napier J, Rouchka EC, Mott JL, Klinge CM. Regulation of miR-29b-1/a transcription and identification of target mRNAs in CHO-K1 cells. Mol Cell Endocrinol. 2017 Mar 15;444:38-47. doi: 10.1016/j.mce.2017.01.044. Epub 2017 Jan 28. PMID: 28137615; PMCID: PMC5316361.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 30: Determining the impact of IgG1 monoclonal antibody production in CHO cells

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will reanalyse RNA-seq data from an IgG1-producing CHO cell line versus its non-producing host. The students will use updated RNA-seq workflows in Galaxy to refine differential expression and pathway analyses.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology. Rives D, Peak C, Blenner MA. RNASeq highlights ATF6 pathway regulators for CHO cell engineering with different impacts of ATF6β and WFS1 knockdown on fed-batch production of IgG1. Sci Rep. 2024 Jun 19;14(1):14141. doi: 10.1038/s41598-024-64767-1. PMID: 38898154; PMCID: PMC11187196.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 31: Identification of transcriptome alteration associated with EPO-Fc production in CHO cells

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will perform an analysis for EPO-Fc–producing CHO cell line relative to its matched host. The students will utilise updated bioinformatics pipelines, to discover new knowledge on how different recombinant proteins influence CHO cell behaviour. Rives D, Peak C, Blenner MA. RNASeq highlights ATF6 pathway regulators for CHO cell engineering with different impacts of ATF6β and WFS1 knockdown on fed-batch production of IgG1. Sci Rep. 2024 Jun 19;14(1):14141. doi: 10.1038/s41598-024-64767-1. PMID: 38898154; PMCID: PMC11187196.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 32: What impact do different media formulations have on CHO cell factories?

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will reanalyse RNA-seq data from CHO cells cultured in two different media, to determine how media composition influences mRNA transcription. Tanemura H, Masuda K, Okumura T, Takagi E, Kajihara D, Kakihara H, Nonaka K, Ushioda R. Development of a stable antibody production system utilizing an Hspa5 promoter in CHO cells. Sci Rep. 2022 May 24;12(1):7239. doi: 10.1038/s41598-022-11342-1. PMID: 35610229; PMCID: PMC9130236.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 33: Comparing the transcriptomes of CHO cell pools vs clonal populations

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare the transcriptomes of CHO cell pools and derived clonal populations, with the goal of defining how clonal selection reshapes gene expression programmes relevant to growth, productivity, stress responses, and product quality attributes in bioprocessing. Students will generate and analyse differential expression and pathway signatures between pools and clones using Galaxy-based pipelines, evaluate transcriptional heterogeneity and stability, and identify candidate genes or pathways that may explain performance differences and guide more rational CHO cell line development strategies. Vito D, Eriksen JC, Skjødt C, Weilguny D, Rasmussen SK, Smales CM. Defining lncRNAs Correlated with CHO Cell Growth and IgG Productivity by RNA-Seq. iScience. 2020 Jan 24;23(1):100785. doi: 10.1016/j.isci.2019.100785. Epub 2019 Dec 18. PMID: 31962234; PMCID: PMC6971398.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 34: Identification of mRNAs associated with lactate production during CHO cell culture

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq data to compare CHO cell lines exhibiting high versus low lactate phenotypes during fed-batch culture, aiming to define transcriptomic signatures and metabolic pathways associated with altered lactate production and consumption, including genes in glycolysis, the TCA cycle, pyruvate metabolism, and lactate transport/regulation. Vito D, Eriksen JC, Skjødt C, Weilguny D, Rasmussen SK, Smales CM. Defining lncRNAs Correlated with CHO Cell Growth and IgG Productivity by RNA-Seq. iScience. 2020 Jan 24;23(1):100785. doi: 10.1016/j.isci.2019.100785. Epub 2019 Dec 18. PMID: 31962234; PMCID: PMC6971398.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 35: Understanding the transcriptional basis for rapid CHO cell growth rate

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare CHO cell lines exhibiting divergent specific growth rates, with the goal of defining transcriptomic signatures and pathways associated with the fast- versus slow-growth phenotype. Vito D, Eriksen JC, Skjødt C, Weilguny D, Rasmussen SK, Smales CM. Defining lncRNAs Correlated with CHO Cell Growth and IgG Productivity by RNA-Seq. iScience. 2020 Jan 24;23(1):100785. doi: 10.1016/j.isci.2019.100785. Epub 2019 Dec 18. PMID: 31962234; PMCID: PMC6971398.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 36: How does the biological system of emerging Chinese hamster cell lines compare to traditional CHO cells

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare IgG1-producing CHO and CHL cell lines, aiming to identify gene expression programmes and pathways that underlie differences in growth, secretion capacity, stress responses, and overall IgG1 productivity between the two host systems. Students will apply standardised Galaxy-based workflows for quality control, alignment, differential expression, and pathway enrichment to define host-specific signatures, highlight candidate engineering targets, and generate hypotheses on which cell line offers the most favourable molecular environment for robust IgG1 manufacture. Tsunoda Y, Arishima R, Boronina T, Cole R, Yamano-Adachi N, Betenbaugh M, Omasa T. Comparative omics profiling reveals differences in biomass, energy production, and vesicle transport between CHO and fast-growing CHL-YN cells. Sci Rep. 2025 Nov 13;15(1):39784. doi: 10.1038/s41598-025-23503-z. PMID: 41233401; PMCID: PMC12615816.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 37: How does reducing cell culture temperature to 33 °C affect the CHO cell transcriptome?

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare CHO cells cultured at 37°C versus 33°C, aiming to define how mild hypothermia reshapes global gene expression programmes controlling cell growth, cell cycle progression, metabolism, stress responses, and recombinant protein synthesis. The student will apply standardised Galaxy-based workflows for quality control, alignment, differential expression, and pathway analysis to link temperature-dependent transcriptomic changes to measured culture performance, generating candidate markers and engineering or process targets to rationally exploit temperature shifts for improved CHO cell bioprocessing outcomes. Nguyen LN, Novak N, Baumann M, Koehn J, Borth N. Bioinformatic Identification of Chinese Hamster Ovary (CHO) Cold-Shock Genes and Biological Evidence of their Cold-Inducible Promoters. Biotechnol J. 2020 Mar;15(3):e1900359. doi: 10.1002/biot.201900359. Epub 2019 Dec 11. PMID: 31785035.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 38: What is the biological basis of clonal variation in IgG1 production rates?

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare two IgG1-producing CHO cell clones that differ in performance, aiming to define clone-specific transcriptional programmes linked to production. Students will apply a Galaxy-based analysis pipeline for quality control, alignment, differential expression, and pathway enrichment to identify genes and pathways that distinguish the two clones, generating candidate molecular markers and engineering targets to explain and potentially exploit their IgG1 productivity differences. Tanemura H, Masuda K, Okumura T, Takagi E, Kajihara D, Kakihara H, Nonaka K, Ushioda R. Development of a stable antibody production system utilizing an Hspa5 promoter in CHO cells. Sci Rep. 2022 May 24;12(1):7239. doi: 10.1038/s41598-022-11342-1. PMID: 35610229; PMCID: PMC9130236.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 39: Comparing mRNA expression between the exponential and stationary phases for CHO-S cells

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare CHO cells harvested in exponential versus stationary phase, aiming to define how the transition from active proliferation to growth arrest reshapes global gene expression in pathways controlling cell cycle, metabolism, stress responses, and recombinant protein synthesis. The students will apply a standardised analysis workflow to identify phase-specific differential expression and pathway enrichment signatures, linking them to culture performance metrics and generating hypotheses on how controlling or prolonging particular phases could improve CHO bioprocess robustness and productivity. Hefzi H *et al*. A Consensus Genome-scale Reconstruction of Chinese Hamster Ovary Cell Metabolism. Cell Syst. 2016 Nov 23;3(5):434-443.e8. doi: 10.1016/j.cels.2016.10.020. PMID: 27883890; PMCID: PMC5132346.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 40: Comparing high and low producing CHO cells

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare high- versus low-producing CHO cell lines expressing the same mAb, with the goal of identifying gene expression programmes and pathways associated with enhanced specific productivity, secretion capacity, and bioprocess robustness. Orellana CA, Marcellin E, Palfreyman RW, Munro TP, Gray PP, Nielsen LK. RNA-Seq Highlights High Clonal Variation in Monoclonal Antibody Producing CHO Cells. Biotechnol J. 2018 Mar;13(3):e1700231. doi: 10.1002/biot.201700231. Epub 2018 Feb 1. PMID: 29316330.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 41: Do subclones derived from existing cell lines have a different transcriptome?

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare high- versus low-producing subclones derived from the same parental CHO line, aiming to identify transcriptional changes that emerge during subcloning and are associated with altered specific productivity, growth, and secretory pathway capacity. Students will apply a common analysis workflow for differential expression and pathway enrichment to define subclone-specific gene expression signatures, generating candidate molecular markers and engineering targets that explain productivity divergence within a single clonal lineage and inform more rational subclone selection strategies. Orellana CA, Marcellin E, Palfreyman RW, Munro TP, Gray PP, Nielsen LK. RNA-Seq Highlights High Clonal Variation in Monoclonal Antibody Producing CHO Cells. Biotechnol J. 2018 Mar;13(3):e1700231. doi: 10.1002/biot.201700231. Epub 2018 Feb 1. PMID: 29316330.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 42: How does reducing cell culture temperature to 31 °C affect the CHO cell transcriptome?

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare CHO cells cultured at 37°C versus 31°C, aiming to characterise how stronger hypothermia reshapes global gene expression programmes controlling cell cycle progression, metabolism, stress responses, and recombinant protein synthesis in intensified or extended cultures. The student will apply a standardised analysis workflow to identify temperature-specific differential expression and pathway enrichment signatures, linking them to observed changes in growth, viability, and productivity, and generating hypotheses on how deep temperature shifts can be exploited to optimise CHO bioprocess performance. Tzani I, Monger C, Motheramgari K, Gallagher C, Hagan R, Kelly P, Costello A, Meiller J, Floris P, Zhang L, Clynes M, Bones J, Barron N, Clarke C. Subphysiological temperature induces pervasive alternative splicing in Chinese hamster ovary cells. Biotechnol Bioeng. 2020 Aug;117(8):2489-2503. doi: 10.1002/bit.27365. Epub 2020 Jun 4. PMID: 32346860.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 43: Comparison of cell lines with different levels of aggregation of bispecific antibody (bsAb1)

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare bispecific antibody–producing CHO cell line that differ in product aggregation (high- versus low-aggregating phenotypes), with the aim of identifying transcriptional changes associated with reduced aggregation. The student students will apply standardised Galaxy-based workflows for differential expression and pathway analysis, integrating aggregation and culture performance data to nominate candidate molecular markers and engineering targets that could be exploited to minimise bispecific antibody aggregation in CHO-based bioprocesses. Barzadd MM et al. Autophagy and intracellular product degradation genes identified by systems biology analysis reduce aggregation of bispecific antibody in CHO cells. N Biotechnol. 2022 May 25;68:68-76. doi: 10.1016/j.nbt.2022.01.010. Epub 2022 Feb 2. PMID: 35123066.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 44: Comparison of cell lines with different levels of aggregation of bispecific antibody (bsAb2)

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare bispecific antibody–producing CHO cell line that differ in product aggregation (high- versus low-aggregating phenotypes), with the aim of identifying transcriptional changes associated with reduced aggregation. The student students will apply standardised Galaxy-based workflows for differential expression and pathway analysis, integrating aggregation and culture performance data to nominate candidate molecular markers and engineering targets that could be exploited to minimise bispecific antibody aggregation in CHO-based bioprocesses. Barzadd MM et al. Autophagy and intracellular product degradation genes identified by systems biology analysis reduce aggregation of bispecific antibody in CHO cells. N Biotechnol. 2022 May 25;68:68-76. doi: 10.1016/j.nbt.2022.01.010. Epub 2022 Feb 2. PMID: 35123066.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 45: Comparison of bispecific antibody producing cell line to the original host cell line (bsAb1)

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare a bispecific antibody–producing CHO cell line to its non-producing host, aiming to identify how expression of the complex bispecific molecule remodels global gene expression in pathways related to growth, metabolism, ER stress, protein folding, secretion, and quality control. The students will apply a standardised Galaxy-based workflow for quality control, alignment, differential expression, and pathway analysis to define producer-specific transcriptional signatures and nominate candidate engineering targets that could enhance bispecific antibody productivity while minimising cellular stress and aggregation. Barzadd MM et al. Autophagy and intracellular product degradation genes identified by systems biology analysis reduce aggregation of bispecific antibody in CHO cells. N Biotechnol. 2022 May 25;68:68-76. doi: 10.1016/j.nbt.2022.01.010. Epub 2022 Feb 2. PMID: 35123066.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 46: Comparison of bispecific antibody producing cell line to the original host cell line (bsAb2)

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare a bispecific antibody–producing CHO cell line to its non-producing host, aiming to identify how expression of the complex bispecific molecule remodels global gene expression in pathways related to growth, metabolism, ER stress, protein folding, secretion, and quality control. The students will apply a standardised Galaxy-based workflow for quality control, alignment, differential expression, and pathway analysis to define producer-specific transcriptional signatures and nominate candidate engineering targets that could enhance bispecific antibody productivity while minimising cellular stress and aggregation. Barzadd MM et al. Autophagy and intracellular product degradation genes identified by systems biology analysis reduce aggregation of bispecific antibody in CHO cells. N Biotechnol. 2022 May 25;68:68-76. doi: 10.1016/j.nbt.2022.01.010. Epub 2022 Feb 2. PMID: 35123066.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.

Project 47: Understanding clonal variation in therapeutic sulfatase in high cell density cultures

Principal Investigator:	Colin Clarke
Laboratory Location:	NIBRT
Modality	Recombinant protein
Focus Area	*Upstream*
Project Description	This project will use RNA-seq to compare sulfatase-producing CHO cell clones that display distinct activity and productivity profiles in high cell density culture, with the aim of identifying clonal transcriptional differences in protein folding, post-translational activation, secretion, and stress-response pathways that underlie variation in active sulfatase yield. The students will apply a time-resolved, Galaxy-based analysis workflow across key stages of the high cell density process to define genes and pathways associated with high specific activity and robust secretion, generating candidate co-expression or engineering targets to enhance sulfatase production in intensified CHO bioprocesses. Thalén NB, et al. Tuning of CHO secretional machinery improve activity of secreted therapeutic sulfatase 150-fold. Metab Eng. 2024 Jan;81:157-166. doi: 10.1016/j.ymben.2023.12.003. Epub 2023 Dec 9. PMID: 38081506.
Anticipated project outcomes	Insights into gene expression patterns revealing novel biological trends or regulatory mechanisms not captured in the original studies, contributing to deeper understanding of CHO cell biology.
Indicative skills and experience to be gained	Students will develop hands-on proficiency in RNA-seq data quality assessment, alignment, and differential expression analysis using the Galaxy platform (https://usegalaxy.eu/). They will also strengthen their ability to interpret transcriptomic data, apply current bioinformatics workflows, and communicate reproducible analytical results within a bioprocessing research context. **Note I:** This project is entirely computational. The will be no need to code, however if the student is interested the project can be modified accordingly. **Note II:** Students undertaking the this project will also attend a **Omics Analysis of Biopharmaceutical Cell Factories** summer school in May/June. The programme will provide foundational understanding of proteomics, transcriptomics, and RNA-seq, equipping participants with the broader biological and analytical context needed to effectively interpret their cell data using modern bioinformatics workflows.